

Interpretable Geometric Analysis of the Humerus in Fat-Suppressed Shoulder MRI

Toward localization of rotator cuff tears via foundation-model segmentation
and direction-aware geometric features

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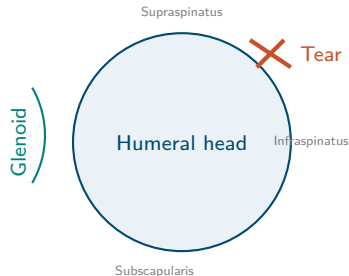
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Bridging computational geometry and musculoskeletal radiology

The clinical problem: rotator cuff tears

Why it matters

- Rotator cuff disease affects **>30% of adults over 60**, the leading cause of shoulder pain and disability.
- Tears arise from acute trauma or chronic degeneration (microtrauma + tendon-insertion ischemia).
- Accurate **localization** — which tendon, where along the insertion — drives surgical planning.



Axial schematic of the glenohumeral joint.
Tears disrupt the articular contour focally.

Why is this hard to read on MRI?

Diagnostic challenge

- MRI reading is **expert-dependent** and time-consuming.
- Inter-vendor variability (SIEMENS / GE / Philips) complicates automation.
- Subtle partial tears can be missed on a single slice; 3D context is needed.

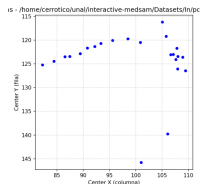
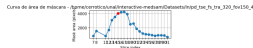
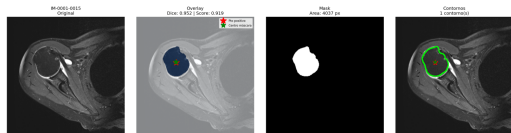
Equity-of-access gap

- Limited access to subspecialty musculoskeletal radiology in low-resource settings.
- In Colombia, regions like **Chocó** (Quibdó) lack dedicated MSK readers despite having modern MRI.
- Automated, interpretable tools can support generalist radiologists.

Why fat-suppressed MRI?

Fat saturation (fs)

- Selective RF pulse **nulls fat signal** $\Rightarrow$ subcutaneous and marrow fat appear dark.
- Cortical bone, tendon, edema and joint fluid become **conspicuous**.
- Standard sequences here: **PD-fs** (anatomy + edema sensitivity) and **T1-fs**.



Axial PD-fs slice with humeral segmentation overlay
(PixelSpacing ≈ 0.47 mm; matrix 320×320).

Why the axial plane?

Geometric structure of the axial stack

- The humeral head appears as **nested circles** that grow to a maximum (the “equator”) and shrink.
- This yields a clean *area-vs-slice* curve that we can validate geometrically.
- The axial plane gives a direct view of the **glenohumeral contact surface**, where most rotator-cuff tears manifest.

Why this matters for the pipeline

The bell-shaped area curve is the **anchor** for downstream validation: trimming non-humeral tails, detecting outlier slices, and selecting the articular slices on which we fit per-slice circles.

From anatomy to geometry: two natural primitives

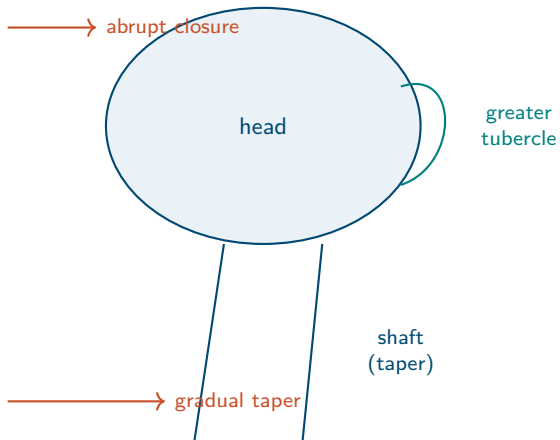
The humeral head $\approx$ spherical cap

- Articular cartilage covers a near-perfect spherical surface (radius $\sim 23\text{--}26$ mm in adults).
- Motivates a **global sphere fit** as a patient-specific anatomical reference.

The articular surface $\approx$ local circles

- In equatorial axial slices, the contact contour with the glenoid is locally circular.
- Motivates **per-slice circles** that *lie on* the articular surface, not approximate it.

Anatomical schematic of the proximal humerus



Sagittal sketch: humeral head, greater tubercle, shaft.

Anatomical asymmetry: head vs shaft

The two ends of the humerus look different

- **Head:** closes **abruptly** at the anatomical neck.
- **Shaft:** tapers **gradually** (longer, smoother).

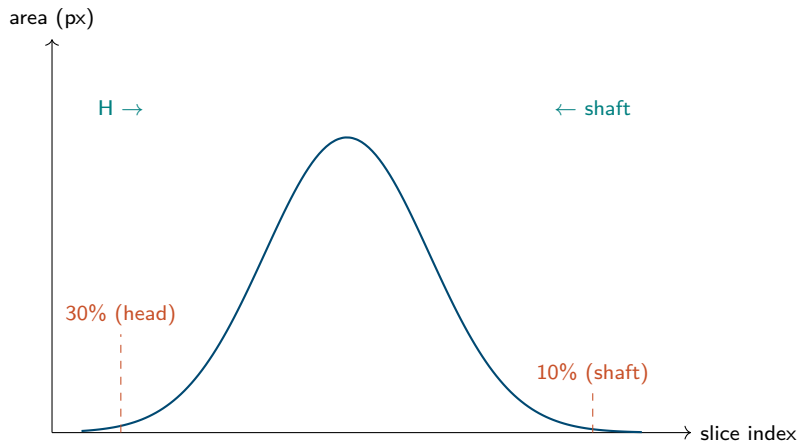
Direct consequence on the pipeline

Two asymmetric thresholds in the area-curve validation:

- **30%** of peak on the **head** side.
- **10%** of peak on the **shaft** side.

Anatomy, not arbitrary heuristics.

Asymmetric trim of the area curve



Asymmetric thresholds: stricter on the head side (abrupt closure), looser on the shaft side (gradual taper).

Hypothesis and objective

Hypothesis

A rotator cuff tear produces a **focal, geometrically detectable** discontinuity of the humeral articular contour. If we extract clinically interpretable geometric features from MRI, we can **localize** the lesion to a specific tendon insertion sector — without training a new neural network.

Objective

Build an **end-to-end pipeline** that:

- ① Segments the humerus across the entire axial volume from a **single seed slice**.
- ② Extracts two interpretable features: **(1)** a global humeral-head **sphere**; **(2)** per-slice articular **circles** that lie on the cartilage surface.
- ③ Uses these features as inputs for downstream tear-localization.

Design principles

Interpretable

Every output is geometrically meaningful and **visually verifiable** by a radiologist: masks, boundary CSVs, fitted circles, summary figures.

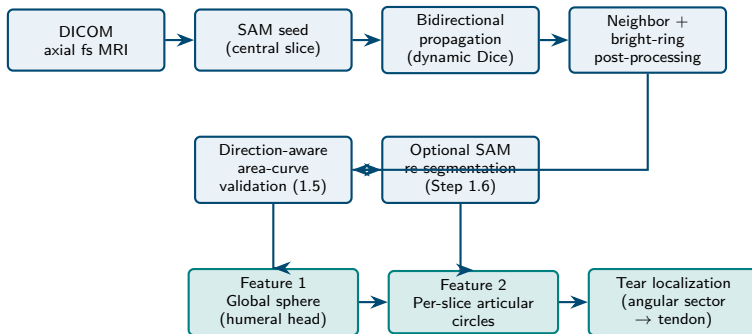
Vendor-agnostic

Works on SIEMENS, GE and Philips data **without retraining**: thresholds are geometric (mm, % of peak), not intensity-based.

Anatomy-aware

Pipeline parameters reflect humeral anatomy (head vs shaft asymmetry, articular surface localization), **not arbitrary heuristics**.

Pipeline overview



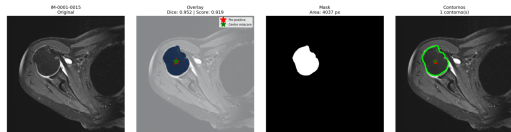
Foundation model (SAM, frozen) + classical geometry + clinical anatomy.

Each block produces auditable artifacts: masks, CSVs, validation reports, summary figures.

Stage 1: SAM segmentation + bidirectional propagation

Foundation-model segmentation

- **SAM (ViT-B)** as a frozen segmenter — no training, no labels.
- One seed slice (click *or* automatic point at $0.40W, 0.40H$).
- Morphological refinement: small-object removal, hole filling, opening/closing.

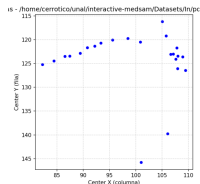
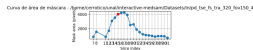


Bidirectional propagation, dynamic Dice

Centroid of slice $i \rightarrow$ positive prompt for $i \pm 1$. **Adaptive threshold:**

$$\tau_i = 0.45 + 0.25 \cdot |i - i_{\text{seed}}| / \Delta_{\text{max}}$$

Strict near the seed, permissive at extremes. **Negative-prompt** fallback for failures.



Per-slice segmentation overlay.

Green = mask boundary;

red dot = positive prompt centroid.

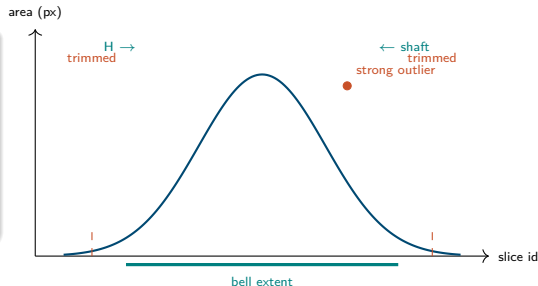
Stage 2: direction-aware validation of the area curve

Reading anatomy from the DICOM header

- Compute slice normal $\mathbf{n} = \text{row} \times \text{col}$ from `ImageOrientationPatient`.
- Project `ImagePositionPatient` on $\mathbf{n} \Rightarrow$ **which end is the head**.
- Fallback: `SliceLocation` $\rightarrow$ `InstanceNumber`.

Why the headers, not conventions

Vendor conventions about scan direction are inconsistent. Reading the geometric headers makes the pipeline **vendor-agnostic by construction**.



Per-slice mask area along the axial stack.

Asymmetric trim and outlier flagging

Asymmetric static floor (anatomy-driven)

- 30% of peak on the head side — abrupt closure at the anatomical neck.
- 10% of peak on the shaft side — gradual taper.

Reversal-from-peak

Walk outward from the peak; trim when the curve genuinely rises again ($\geq 10\%$ on the head side, $\geq 20\%$ on the shaft side, with a ≥ 50 px absolute floor).

Internal outlier flagging

Robust MAD residual ($\geq 2.5\sigma$), cross-checked with circle inlier-ratio drop ($\geq 15\%$) or Dice drop ($\geq 10\%$). Strong outliers are excluded; weak ones are reported but kept.

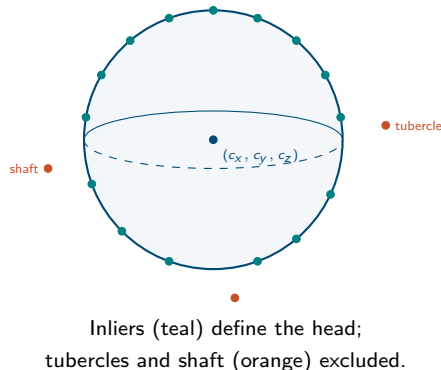
Feature 1: global humeral-head sphere

What it is

A single sphere fitted to the 3D contour cloud (DICOM millimeters): a patient-specific **anatomical reference** for the humeral head.

How it is computed

- Pixel $\rightarrow$ world via IPP, IOP, PixelSpacing.
- **RANSAC sphere**: 2000 iterations, 4-point sampling, Kåsa fit.
- Inlier distance ≤ 2.0 mm; final refit by least-squares on inliers.
- Sanity bound: $r < 200$ mm.



Feature 1: clinical interpretation

Patient-specific anatomical reference

- Recovers the **expected curvature** of the humeral head (typical $r \in 23\text{--}26$ mm in adults).
- Centroid $\mathbf{c}$ + radius r characterize the “ideal” humeral head for this patient.
- Outliers (tubercles, shaft, artifacts) are excluded **automatically** by RANSAC.

Why it matters clinically

- Enables **normalized comparison** across patients, vendors, and acquisitions.
- Provides a per-patient *baseline geometry* against which local deviations stand out.
- Sanity-checks downstream features: a per-slice circle far from the global sphere is suspicious.

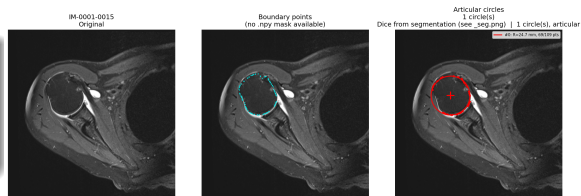
Feature 2: per-slice articular circles

Key idea

These circles do **not approximate** the sphere — they **lie on** the cartilage surface where the humeral head meets the glenoid.

How they are computed

- Articular slices: mask area $\geq 70\%$ of the peak.
- Project boundary to the slice's best-fit plane via **SVD**.
- Iterative 2D RANSAC + Kåsa fit (up to 3 circles per slice).
- Inlier distance ≤ 1.5 mm; ratio $\geq 35\%$; min. 10 inliers.



Five-panel render for one articular slice.
The fitted circle lies on the articular contour.

Feature 2: stored information and clinical meaning

What each circle stores

- Center in 3D millimeters and radius.
- In-plane orthonormal basis ($\mathbf{u}, \mathbf{v}$) and slice normal $\mathbf{n}$.
- Inlier point cloud and residual statistics.
- Re-projection on the original axial PNG for slice-by-slice clinician review.

Clinical interpretation

- A focal **drop in inlier ratio** on a circle indicates loss of articular continuity.
- A sudden **shift of the center** or radius vs neighbors flags asymmetric pathology.
- Each circle is rendered as a `*_circle.png` so the radiologist can verify each fit.

The geometric trick: parameterizing the articular circle

From a circle to a 1D angular map

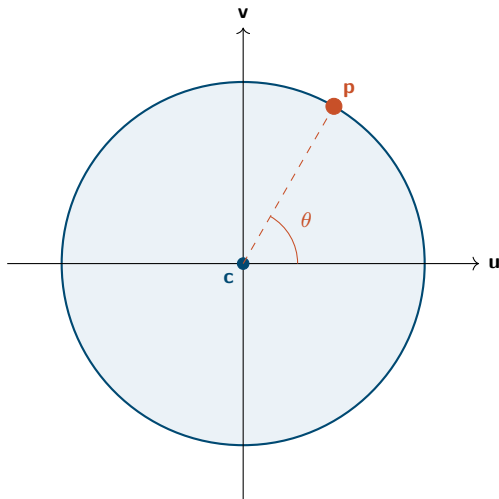
Each per-slice circle has an in-plane orthonormal basis $(\mathbf{u}, \mathbf{v})$ and center $\mathbf{c}$. Any inlier point $\mathbf{p}$ has angular coordinate

$$\theta = \text{atan2}((\mathbf{p} - \mathbf{c}) \cdot \mathbf{v}, (\mathbf{p} - \mathbf{c}) \cdot \mathbf{u}).$$

Why this is powerful

The articular circle becomes a **1D tendon map**: each θ on the circle corresponds to a known anatomical direction at the humeral head, the same for every patient and every vendor.

Angular parameterization of the circle



θ is the angular coordinate of every inlier on the articular circle.

Angular sectors → rotator-cuff tendons

Anatomical sector mapping

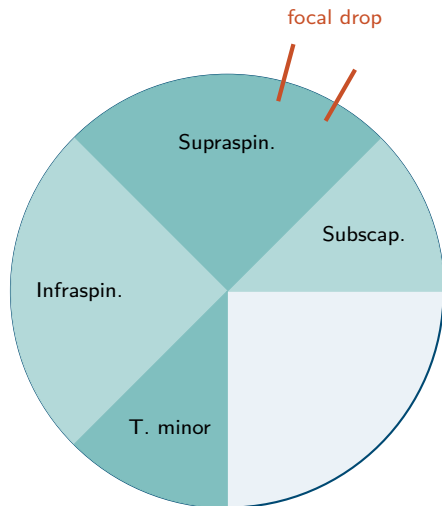
Tendon insertions fall in **predictable angular sectors** on the articular circle:

- **Subscapularis** → lesser tubercle: $\sim 0-45^\circ$.
- **Supraspinatus** → greater tub. (sup): $\sim 45-135^\circ$.
- **Infraspinatus** → greater tub. (post): $\sim 135-225^\circ$.
- **Teres minor** → greater tub. (inf-post): $\sim 225-270^\circ$.

Localization rule

A focal drop in inlier ratio in a specific sector → **candidate tear** in the corresponding tendon, at the slice height where the drop occurs.

Sector diagram: tendon insertions and a focal drop



The articular circle becomes a 1D *tendon map*; a focal drop in inlier ratio identifies the affected tendon.

Multi-vendor robustness and equity of access

Tested across vendors and sites

- **SIEMENS** – Quibdó, PD-fs TSE.
- **GE** – Optima MR360, Bogotá, PD-fs FSE.
- **Philips** – Ingenia, SOMA, ePDW SPAIR.
- Matrix 256×256 to 320×320 ; PD-fs and T1-fs.

Dataset	Kept	Articular
SIEMENS P5SE1 (Quibdó)	21/32	5
SIEMENS PD fs (fov150)	18/23	4
SIEMENS T1 fs #5	19/24	5
SIEMENS T1 fs #11	17/22	4
GE Optima MR360 (Bogotá)	16/21	4
Philips Ingenia (SOMA)	20/26	5

Why this generalizes

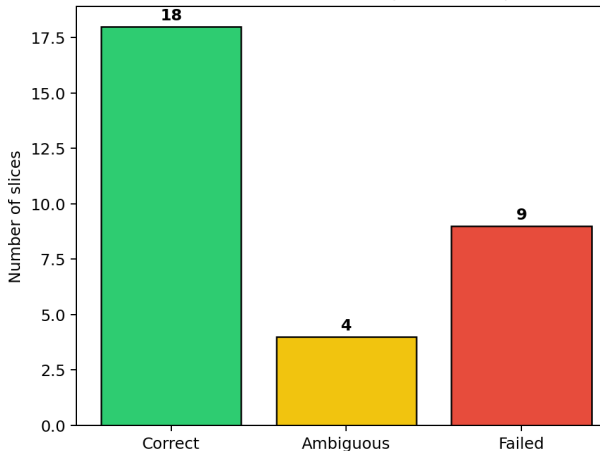
- Frozen SAM $\Rightarrow$ no domain-specific weights to overfit.
- Thresholds are **geometric** (mm, % of peak), not intensity-based.
- Direction handling reads DICOM headers, not vendor conventions.

Equity-of-access angle

Datasets include **Quibdó**, **Chocó**, a region with limited subspecialty radiology. Vendor-agnostic, interpretable AI directly supports diagnostic capacity in such settings.

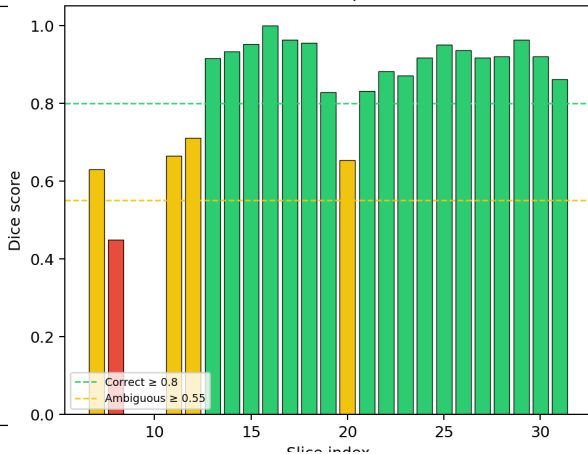
Qualitative results — slice quality (1/3)

1. Slice Classification (Dice-based)



Slice classification by Dice.

2. Dice Score per Slice

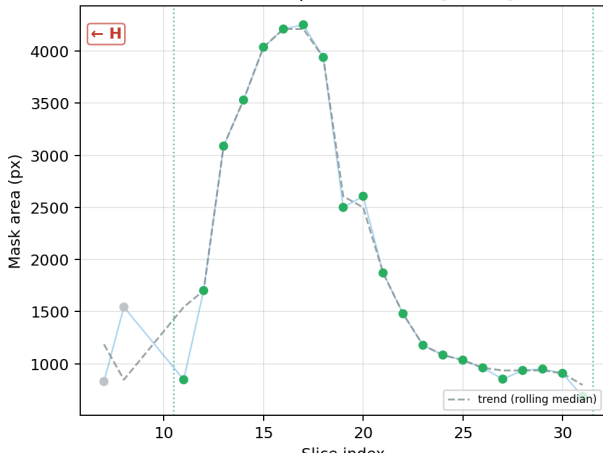


Dice score per slice.

Most slices land in **Correct**; failures cluster at the head/shaft extremes.

Qualitative results — area curve and 3D boundary (2/3)

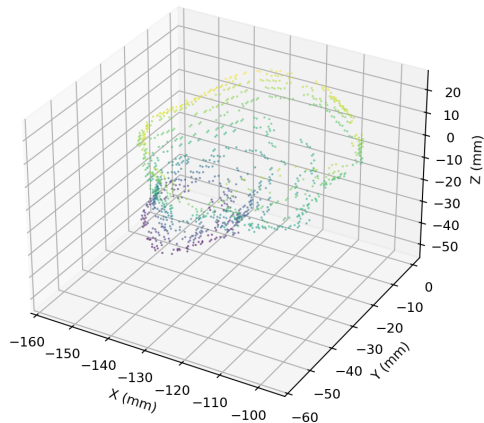
3. Mask Area per Slice bell=[11..31]



Mask area per slice with bell extent (Step 1.5).

H→ indicates the head side. Trimmed tails appear in gray; outliers in red.

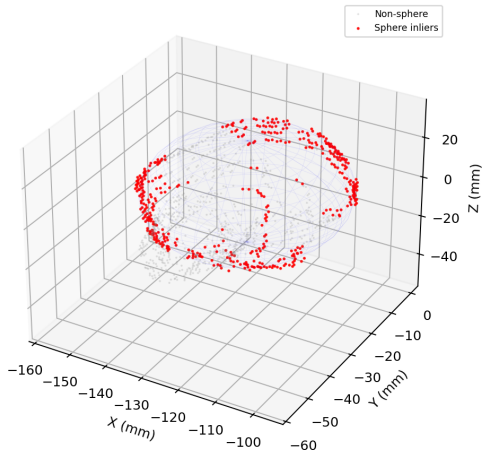
4. Boundary Points (DICOM coords)



Boundary points in DICOM mm coordinates.

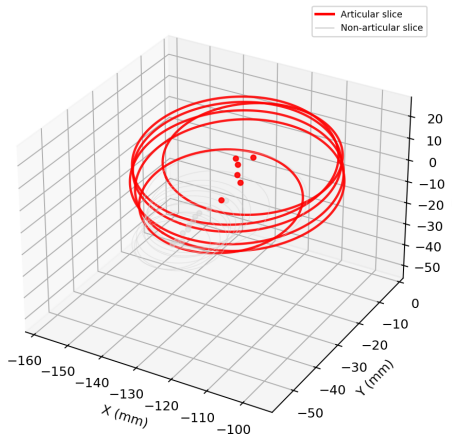
Qualitative results — fitted geometry (3/3)

5. Sphere Fit $R=25.2$ mm



Feature 1: humeral-head sphere ($R = 25.2$ mm).
17 circles fitted; 6 of them lie on the articular surface.

6. Per-slice Circles (Feature 2)
17 circles, 6 on articular surface



Feature 2: per-slice articular circles.

How to read these figures

Color code (area curve)

- **Green**: slices kept for analysis.
- **Gray**: tails trimmed by direction-aware logic.
- **Red**: strong outliers excluded after Pass 2.
- Vertical lines: bell-extent boundaries.
- Caret H→: which side of the volume is the head.

Auditability by design

Every slice exports a five-panel *_circle.png (image, mask, contour, fitted circle, overlay). A radiologist can verify each fit **slice by slice**, not just trust an aggregate score.

Reproducibility

Thresholds are CLI flags; every run writes area_curve_validation.csv for downstream audit.

Limitations and clinical considerations

Current limitations

- **Validation cohort size:** 6 datasets across 3 vendors — proof of concept, not clinical validation.
- **Seed-slice selection:** automatic point at $(0.40W, 0.40H)$ assumes typical centering; pathological positioning may fail.
- **Pattern B without GPU:** when SAM grabs adjacent muscle/glenoid, weak outliers are flagged but not re-segmented unless GPU + SAM are available (Step 1.6).
- **Tear-localization rule is a hypothesis:** the angular-sector mapping needs prospective validation against arthroscopic ground truth.

What is required for clinical translation

- **Multi-center, larger cohort** with arthroscopy or surgical confirmation.
- Inter-reader study: do radiologists agree with the geometric flag location?
- Integration with PACS / DICOM-SR for routine workflow.
- Reproducibility: containerized pipeline, public benchmarks.

Risk profile

Because outputs are **geometric, visualized, and auditable**, the system fails “loudly” rather than silently — a key property for clinical deployment compared to opaque end-to-end models.

What we have built

- A **frozen foundation model** (SAM) plus **anatomy-aware classical geometry** produces interpretable shoulder MRI features without training.
- DICOM-header reading makes the pipeline **direction-aware** with asymmetric anatomy-driven thresholds.
- Two features — humeral sphere + per-slice articular circles — enable **tear localization by angular sector**.
- Robust across SIEMENS, GE and Philips; relevant for low-resource radiology settings.

Next steps and acknowledgements

Next steps

- Implement the angular-sector → tendon mapping end-to-end.
- Prospective validation against arthroscopy.
- Inter-reader and inter-vendor reproducibility study.
- Integration into a clinical reading viewer.

Acknowledgements

- Universidad Nacional de Colombia — Topics in Computational Geometry.
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Thank you — questions welcome.

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